
MATH 562: STATISTICAL INFERENCE

Homework 2

Rafael Barroso

Ingenierie Mathematique
École Polytechnique Fédérale de Lausanne
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1 Preliminaries

The aim for this section is to introduce some statistical notions that will be used in order to solve the exercises in this homework. We begin exploring the concept of a *statistical pivot*. A statistical pivot (or *pivotal quantity*) is a function of both the sample data and one or more unknown parameters whose **probability distribution does not depend on those parameters**.

Definition 1.1 (Statistical Pivot). Formally, if $X = (X_1, \dots, X_n)$ is a random sample from a distribution depending on a parameter θ , a statistic $T(X, \theta)$ is called a *pivot* if the distribution of T is the same for all possible values of θ .

In simpler terms, a pivot is a clever combination of our data X and the unknown parameters θ that “cancels out” the dependence on those parameters, leaving something whose behavior is fully known. As we’ll explore throughout the worked exercises, pivotal quantities are extremely useful in statistical inference because they allow us to construct **confidence intervals** and perform **hypothesis tests** without knowing the true value of the underlying parameter.

Remark 1.2. By using a pivot, we can make probability statements that remain valid for any possible value of θ .

We now explore the concept of *equivariant estimators*. Equivariant estimators are statistical estimators that maintain a consistent relationship with the data under certain transformations.

Definition 1.3 (Equivariant estimators). A parameter estimator $g(y)$ is said to be **equivariant** if it satisfies:

$$g(a + by) = g(a + by_1, \dots, a + by_n) = a + g(by_1, \dots, by_n) = a + bg(y)$$

for all $a \in k$ and $b > 0$.

It is useful to recall the following remark regarding the equivariance properties of estimators in a location-scale model.

Remark 1.4. For estimators in a location-scale model

$$Y_i = \eta + \tau V_i, \quad i = 1, \dots, n,$$

it is natural to require that they behave consistently under changes of location and scale. A **location estimator** $m(\cdot)$ is said to be *location-scale equivariant* if

$$m(\eta + \tau v_1, \dots, \eta + \tau v_n) = \eta + \tau m(v_1, \dots, v_n),$$

that is, translating and rescaling the data simply translates and rescales the estimator in the same way. Similarly, a **scale estimator** $s(\cdot)$ is said to be *location invariant* and *scale equivariant* if

$$s(\eta + \tau v_1, \dots, \eta + \tau v_n) = |\tau| s(v_1, \dots, v_n).$$

These properties ensure that both estimators respond to shifts and rescalings of the data, preserving the structure of the underlying model.

We now introduce the notion of quantiles, which will be useful in the exercises.

Definition 1.5 (Quantiles). For a random variable X with cumulative distribution function (CDF) $F_X(x)$, the quantile q_p is the value such that:

$$\mathbb{P}(X \leq q_p) = p$$

i.g. $q_{0.5}$ is the median of X . This is just a fancy way of saying ‘the point where the CDF hits a certain probability’.

We briefly recall the definition of median and interquartile range (IQR), which will be used in the exercises.

Definition 1.6 (Median). The median of a dataset is the **middle value** when the data points are arranged in ascending order. Thus, it may be thought of as a quantile $q_{0.5}$ such that

$$\mathbb{P}(q_i \leq q_{0.5}) = 0.5$$

- If n is odd, the median is the value at position $\frac{n+1}{2}$.
- If n is even, the median is the average of the values at positions $\frac{n}{2}$ and $\frac{n}{2} + 1$.

Now, we define the interquartile range (IQR) as follows.

Definition 1.7 (Interquartile Range). The interquartile range (IQR) is a measure of statistical dispersion, or how spread out the values in a dataset are. It is defined as the difference between the third quartile (Q_3) and the first quartile (Q_1):

$$\text{IQR} = Q_3 - Q_1$$

where:

- Q_1 (the first quartile) is the median of the lower half of the dataset (the 25th percentile i.e. the $q_{.25}$ quantile!).
- Q_3 (the third quartile) is the median of the upper half of the dataset (the 75th percentile or $q_{.75}$ quantile).

Note that percentiles and quantiles are very closely related concepts. Percentiles are specific types of quantiles that divide a dataset into 100 equal parts. Thus, percentiles can be viewed as quantiles expressed in terms of percentages.

2 Exercise

2.1 Defining the problem

A *location-scale* model is of the form $y_0 = \eta + \tau v_0$ where η is the location parameter (a scalar i.e. $\eta \in k$ for an arbitrary field k), $\tau > 0$ is the scale parameter and v_0 is a random variable with known distribution. Given a *random sample* $y = (y_1, \dots, y_n)$ with the y_i 's being independent, identically distributed (*i.i.d*) random variables from the population y_0 , we may produce estimators $m(y), s(y)$ for the parameters η, τ respectively.

- Show that $q_1(y, \eta) = \frac{m(y) - \eta}{s(y)}$ and $q_2(y, \tau) = \frac{s(y)}{\tau}$ are pivots. Moreover, explain how could we construct confidence intervals for the parameters η, τ using these pivots.
- Deduce that pivots can be formed by taking:
 - $m_1(y) = \bar{y}$ and $s_1(y) = \sqrt{\sum (y_i - \bar{y})^2}$.
 - $m_2(y) = \text{median}(y)$ and $s_2(y) = \mathbf{IQR}(y)$ i.e. the interquartile range.
- Provide a pivot that could be used to construct a prediction interval for y_0 based on $y_1, \dots, y_n$.

2.2 Solution

In order to verify that q_1, q_2 are pivots, we need to show that their distributions do not depend on the parameters η, τ . In order to do so, we will use the fact that m, s are equivariant estimators of location and scale respectively. We write:

$$\begin{aligned} q_1(y, \eta) &= \frac{m(y) - \eta}{s(y)} = \frac{m(y_1, \dots, y_n) - \eta}{s(y_1, \dots, y_n)} = \frac{m(\eta + \tau v_1, \dots, \eta + \tau v_n) - \eta}{s(\eta + \tau v_1, \dots, \eta + \tau v_n)} \\ &\Rightarrow q_1(y, \eta) = \frac{\eta + \tau m(v) - \eta}{|\tau|s(v)} = \frac{\tau m(v)}{\tau s(v)} = \frac{m(v)}{s(v)} \end{aligned}$$

hence the distribution of q_1 does not depend on η, τ and is in fact the distribution of $\frac{m(v)}{s(v)}$. Similarly, we have:

$$\begin{aligned} q_2(y, \tau) &= \frac{s(y)}{\tau} = \frac{s(y_1, \dots, y_n)}{\tau} = \frac{s(\eta + \tau v_1, \dots, \eta + \tau v_n)}{\tau} \\ &\Rightarrow q_2(y, \tau) = \frac{|\tau|s(v)}{\tau} = \frac{\tau s(v)}{\tau} = s(v) \end{aligned}$$

so, the distribution of q_2 does not depend on η, τ and is in fact the distribution of $s(v)$.

In order to construct confidence intervals for η, τ using these pivots, we must first choose a confidence level $1 - \alpha$ for some $\alpha \in (0, 1)$. Then, we need to find the quantiles $q_{\alpha_U}, q_{1-\alpha_L}$ of the distribution of q_i .

Remark 2.1. The quantiles $q_{\alpha_U}, q_{1-\alpha_L}$ are the upper and lower 'cutoff' points of the distribution of q_i such that

$$\mathbb{P}(q_i \leq q_{\alpha_U}) = \alpha_U; \quad \mathbb{P}(q_i \leq q_{1-\alpha_L}) = 1 - \alpha_L$$

and so,

$$\mathbb{P}(q_{\alpha_U} \leq q_i \leq q_{1-\alpha_L}) = 1 - (\alpha_U + \alpha_L)$$

where $\alpha_U + \alpha_L = \alpha$.

The following figure 1 illustrates the quantiles $q_{\alpha_U}, q_{1-\alpha_L}$ of the distribution of q_i . Also, the shaded areas denote the probabilities α_U, α_L .

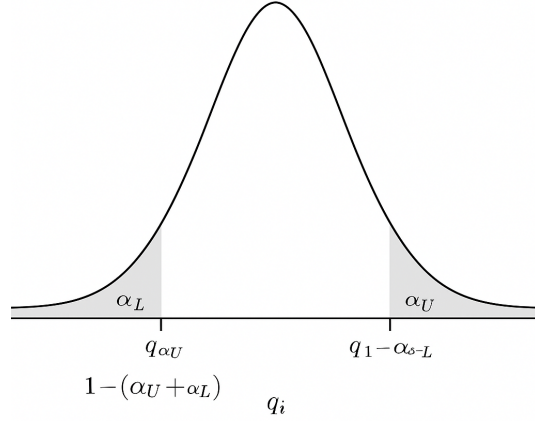


Figure 1: Distribution q_i with quantiles $q_{\alpha_U}, q_{1-\alpha_L}$.

By the above remark, we may transform the probability statement $\mathbb{P}(q_{\alpha_U} \leq q_i \leq q_{1-\alpha_L}) = 1 - \alpha$ into:

$$\mathbb{P}\left(q_{\alpha_U} \leq \frac{m(y) - \eta}{s(y)} \leq q_{1-\alpha_L}\right) = 1 - \alpha$$

$$\mathbb{P}(m(y) - q_{1-\alpha_L} s(y) \leq \eta \leq m(y) - q_{\alpha_U} s(y)) = 1 - \alpha$$

which is a $1 - \alpha$ confidence interval for η . Similarly, we have:

$$\mathbb{P}(q_{\alpha_U} \leq q_2 \leq q_{1-\alpha_L}) = 1 - \alpha$$

$$\mathbb{P}\left(q_{\alpha_U} \leq \frac{s(y)}{\tau} \leq q_{1-\alpha_L}\right) = 1 - \alpha$$

$$\mathbb{P}\left(\frac{s(y)}{q_{1-\alpha_L}} \leq \tau \leq \frac{s(y)}{q_{\alpha_U}}\right) = 1 - \alpha$$

which is a $1 - \alpha$ confidence interval for τ .

Now, we will show that the pairs $(m_1, s_1), (m_2, s_2)$ produce pivots. We start with the first pair:

$$q_1(y, \eta) = \frac{m_1(y) - \eta}{s_1(y)} = \frac{\bar{y} - \eta}{\sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$$

Before proceeding and expanding the equation above like possessed devils, we make the following remark:

Remark 2.2. Note that $\bar{y} = \frac{1}{n} \sum_{i=1}^n (\eta + \tau v_i) = \frac{1}{n} (\eta + \tau v_1 + \dots + \eta + \tau v_n)$. Hence,

$$\bar{y} = \frac{1}{n} (n\eta + \tau (\sum_{i=1}^n v_i)) = \eta + \tau \bar{v}$$

And so, we may write $\sqrt{\sum_{i=1}^n (y_i - \bar{y})^2} = \sqrt{\sum_{i=1}^n (\eta + \tau v_i - (\eta + \tau \bar{v}))^2}$. Thus,

$$\sqrt{\sum_{i=1}^n (y_i - \bar{y})^2} = \tau \sqrt{\sum_{i=1}^n (v_i - \bar{v})^2}$$

By the above remark, we get that

$$q_1(y, \eta) = \frac{\bar{v}}{\sqrt{\sum_{i=1}^n (v_i - \bar{v})^2}}.$$

Note that this only depends on the v_i 's where we know the distribution, making q_1 a pivot. Similarly, we have:

$$\begin{aligned} q_2(y, \tau) &= \frac{s_1(y)}{\tau} = \frac{\sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}{\tau} = \frac{\tau \sqrt{\sum_{i=1}^n (v_i - \bar{v})^2}}{\tau} \\ &\Rightarrow q_2(y, \tau) = \sqrt{\sum_{i=1}^n (v_i - \bar{v})^2} \end{aligned}$$

Once again, this only depends on the v_i 's where we know the distribution, making q_2 a pivot. Using the previous procedure (we might think of it as a recipe for cooking confidence intervals), we may construct confidence intervals for η, τ using these pivots as follows:

$$\begin{aligned} \mathbb{P} \left(\bar{y} - q_{1-\alpha_L} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2} \leq \eta \leq \bar{y} - q_{\alpha_U} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2} \right) &= 1 - \alpha \\ \mathbb{P} \left(\frac{\sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}{q_{1-\alpha_L}} \leq \tau \leq \frac{\sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}{q_{\alpha_U}} \right) &= 1 - \alpha \end{aligned}$$

We may now proceed with the second pair (m_2, s_2) by recalling the following idea; For any $p \in (0, 1)$, empirical p -quantiles are location-scale equivariant (since we're assuming (m_2, s_2) are estimators of location and scale parameters and they're also equivariant by assumption):

$$\hat{q}_p(\eta + \tau v) = \eta + \tau \hat{q}_p(v).$$

Apply this with $p \in \{0.25, 0.5, 0.75\}$:

$$m_2(y) = \hat{q}_{0.5}(y) = \eta + \tau \hat{q}_{0.5}(v), \quad s_2(y) = \hat{q}_{0.75}(y) - \hat{q}_{0.25}(y) = \tau(\hat{q}_{0.75}(v) - \hat{q}_{0.25}(v)).$$

Therefore

$$q_1(y, \eta) = \frac{m_2(y) - \eta}{s_2(y)} = \frac{\eta + \tau \hat{q}_{0.5}(v) - \eta}{\tau(\hat{q}_{0.75}(v) - \hat{q}_{0.25}(v))} = \frac{\hat{q}_{0.5}(v)}{\hat{q}_{0.75}(v) - \hat{q}_{0.25}(v)},$$

which depends only on v and not on (η, τ) . As for q_2 :

$$q_2(y, \tau) = \frac{s_2(y)}{\tau} = \frac{\tau(\hat{q}_{0.75}(v) - \hat{q}_{0.25}(v))}{\tau} = (\hat{q}_{0.75}(v) - \hat{q}_{0.25}(v))$$

again, we're left with a function which only depends on v and not on (η, τ) . Hence, both q_1, q_2 are pivots. Using the previous procedure, we may construct confidence intervals for η, τ using these pivots as follows:

$$\begin{aligned} \mathbb{P}(\text{median}(y) - q_{1-\alpha_L} \text{IQR}(y) \leq \eta \leq \text{median}(y) - q_{\alpha_U} \text{IQR}(y)) &= 1 - \alpha \\ \mathbb{P}\left(\frac{\text{IQR}(y)}{q_{1-\alpha_L}} \leq \tau \leq \frac{\text{IQR}(y)}{q_{\alpha_U}}\right) &= 1 - \alpha \end{aligned}$$

Finally, we provide a pivot that can be used to construct a prediction interval for y_0 based on $y_1, \dots, y_n$. We start by writing:

$$\begin{aligned} q_3(y, y_0) &= \frac{y_0 - m(y)}{s(y)} = \frac{y_0 - m(y_1, \dots, y_n)}{s(y_1, \dots, y_n)} = \frac{y_0 - m(\eta + \tau v_1, \dots, \eta + \tau v_n)}{s(\eta + \tau v_1, \dots, \eta + \tau v_n)} \\ \Rightarrow q_3(y, y_0) &= \frac{\eta + \tau v_0 - (\eta + \tau m(v))}{|\tau|s(v)} = \frac{\tau(v_0 - m(v))}{\tau s(v)} = \frac{v_0 - m(v)}{s(v)} \end{aligned}$$

hence the distribution of q_3 does not depend on η, τ and is in fact the distribution of $\frac{v_0 - m(v)}{s(v)}$, making q_3 a pivot. Using the previous procedure, we may construct a prediction interval for y_0 using this pivot as follows:

$$\mathbb{P}(m(y) + q_{\alpha_U} s(y) \leq y_0 \leq m(y) + q_{1-\alpha_L} s(y)) = 1 - \alpha$$

Remark 2.3. Note that a prediction interval is very similar to a confidence interval. The only difference is the fact that confidence intervals quantify uncertainty for a certain parameter (i.g. η, τ) while prediction intervals quantify uncertainty for a future observation (i.g. y_0).